

ME 333
MECHANICS OF MATERIALS

Lecture 24: Plane Problems in Linear Elasticity II

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Plane Problems in Linear Elasticity
Plane Problems & Stress Method

Procedure Summary :

- Step 1: Use compatibility equation and recast it in stresses using Hooke's Law.
- Step 2: Use equilibrium equations expressed in terms of stress.
- Step 3: Obtain mixed partial derivatives operating on the shear component.
- Step 4: Eliminate shear stress from the compatibility equation derived in Step1 to conclude $\nabla^2(\sigma_{xx} + \sigma_{yy}) = 0$
- Step 5 : Define Airy stress function in terms of stresses as

$$\sigma_{xx} = \phi_{,yy}; \quad \sigma_{yy} = \phi_{,xx}; \quad \tau_{xy} = -\phi_{,xy}$$

Above choice trivially satisfies the equilibrium equations.

- Step 6 : The Laplace equation recast in ϕ results in the *biharmonic equation* given by $\nabla^4\phi = 0$. **Objective :** Determine ϕ .

Plane Problems in Linear Elasticity
Solutions to Biharmonic Equation

- 1 In case of plane static problems when body forces are neglected the governing equation is:

$$\nabla^4\phi = 0; \quad \sigma_{xx} = \frac{\partial^2\phi}{\partial y^2}; \quad \sigma_{yy} = \frac{\partial^2\phi}{\partial x^2}; \quad \sigma_{xy} = \frac{\partial^2\phi}{\partial x\partial y}$$

- 2 Broadly there are two methods of solving the *biharmonic equation* :
 - Inverse Methods: We examine solution and identify BVP corresponding to it.
 - Semi-inverse method: Stress or displacement distribution is assumed.
 - Polynomial method
 - Fourier series method
 - Complex variable method

We will limit the plane problem analysis to *Polynomial Method* only.

Plane Problems in Linear Elasticity
Plane Problems

1 Summary :

- Governing equation irrespective of plane stress or plane strain is:

$$\nabla^4\phi = 0 \quad (\text{biharmonic equation})$$

where: ϕ is the *Airy stress function*.

- The Airy stress function is defined as:

$$\sigma_{xx} = \frac{\partial^2\phi}{\partial y^2} = \phi_{,yy}; \quad \sigma_{yy} = \frac{\partial^2\phi}{\partial x^2} = \phi_{,xx}; \quad \tau_{xy} = -\frac{\partial^2\phi}{\partial x\partial y} = -\phi_{,xy}$$

- **Objective :** Determine ϕ
- **Approach :** Solve the biharmonic equation.

Polynomial Method

Note: Any polynomial of degree 3 or less will satisfy the biharmonic equation.

Some Simple Examples

- [Example 1] : $\phi = Ay^2$ Then:

$$\sigma_{xx} = \phi_{,yy} = 2A; \quad \sigma_{yy} = \phi_{,xx} = 0; \quad \tau_{xy} = -\phi_{,xy} = 0$$

This represents a state of uniaxial tension.

- [Example 2] : $\phi = Bxy$

$$\sigma_{xx} = \sigma_{yy} = 0; \quad \sigma_{xy} = -B$$

This represents a state of pure shear

- [Example 3] : $\phi = Ay^2 + Bxy$.

$$\sigma_{xx} = 2A; \quad \sigma_{yy} = 0; \quad \sigma_{xy} = -B$$

This represents a state that is a superposition of stress states in examples 1 and 2, i.e., a uniaxial stress state superposed with a state of pure shear.



Polynomial Method

Note: Any polynomial of degree 3 or less will satisfy the biharmonic equation

Some Simple Examples (Contd ...)

- [Example 4] : Consider any second order polynomial $\phi = Ax^2 + Bxy + \frac{C}{2}y^2$

$$\sigma_{xx} = C; \quad \sigma_{yy} = 2A; \quad \sigma_{xy} = -B$$

This represents a constant stress state with all stresses being spatially independent.

- [Example 5] : $\phi = Ay^3$

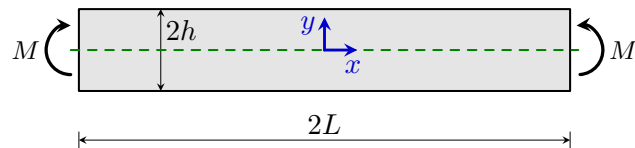
$$\sigma_{xx} = 6Ay; \quad \sigma_{yy} = 0; \quad \sigma_{xy} = 0$$

This represents a case of pure bending as the normal stress in x is dependent on y .



Pure Bending in Beams - Example Problem

[Problem 1] : Derive the displacement, strain and stress fields for a beam under pure bending considering it as a plane problem.



- Step 1 : From the previous example, we can consider $\phi = Ay^3$

$$\sigma_{xx} = \phi_{,yy} = 6Ay; \quad \sigma_{yy} = 0; \quad \sigma_{xy} = 0$$



Pure Bending in Beams - Example Problem

[Problem 1] : Contd ...

- Step 2 : Enforce force and moment equilibrium relations to establish the relation between the stress and applied bending moment

$$\begin{aligned} \int_{-h}^h \sigma_{xx}(x \pm L, y) dy &= 0; \quad \int_{-h}^h \sigma_{xx}(x \pm L, y) y dy = -M; \\ \Rightarrow 2 \int_0^h 6Ay^2 dy &= 4Ah^3 = -M \quad \Rightarrow A = -\frac{M}{4h^3} \end{aligned}$$

- Step 3 : Writing the stress field description in terms of M as

$$\sigma_{xx} = -\frac{3M}{2h^3}y; \quad \sigma_{yy} = \sigma_{xy} = 0$$

Note that $(-3M/2h^3) = -(M/I)$



Pure Bending in Beams - Example Problem

[Problem 1]: Contd ...

- Step 4: Integrate the strain-displacement relations to obtain the displacement field.

$$\begin{aligned}\epsilon_{xx} &= \frac{\partial u}{\partial x} = -\frac{3M}{2Eh^3}y \implies u(x, y) = -\frac{3M}{2Eh^3}xy + f(y) \\ \epsilon_{yy} &= \frac{\partial v}{\partial y} = -\nu\epsilon_{xx} = \nu\frac{3M}{2Eh^3}y \implies v(x, y) = \nu\frac{3M}{2Eh^3}y^2 + g(x) \\ \epsilon_{xy} &= 0 = \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) \implies -\frac{3M}{2Eh^3}x + \frac{\partial f(y)}{\partial y} + \frac{\partial g(x)}{\partial x} = 0\end{aligned}$$

Using separation of variables:

$$\begin{aligned}-\frac{3M}{2Eh^3}x + \frac{\partial g(x)}{\partial x} &= C_1; \quad \frac{\partial f(y)}{\partial y} = -C_1 \\ \implies g(x) &= \frac{3M}{4Eh^3}x^2 + C_1x + C_3; \quad f(y) = -C_1y + C_2\end{aligned}$$

Pure Bending in Beams - Example Problem

[Problem 1]: Contd ...

- Step 5: Substitute $g(x)$ and $f(y)$ in to the displacement fields in Step 4.

$$\begin{aligned}u(x, y) &= -\frac{3M}{2Eh^3}xy - C_1y + C_2 \\ v(x, y) &= \nu\frac{3M}{4Eh^3}y^2 + \frac{3M}{4Eh^3}x^2 + C_1x + C_3\end{aligned}$$

Objective: Determine the unknown coefficients C_1 , C_2 and C_3 .

How?

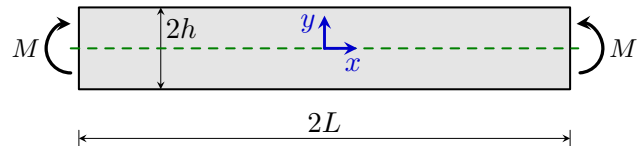
Apply boundary conditions.

Pure Bending in Beams - Example Problem

[Problem 1]: Contd ...

- Step 6a: Identify appropriate boundary conditions.

$u(0, y) = 0$; $v(\pm L, 0) = 0$ Rigid-body motions are fixed



- Step 6b: Imposing boundary conditions.

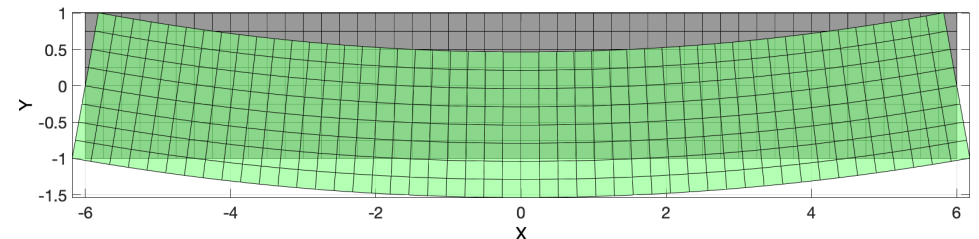
$$u(0, y) = 0 \implies C_1 = C_2 = 0; \quad v(\pm L, 0) = 0 \implies C_3 = -\frac{3ML^2}{4Eh^3}$$

Pure Bending in Beams - Example Problem

[Problem 1]: Contd ...

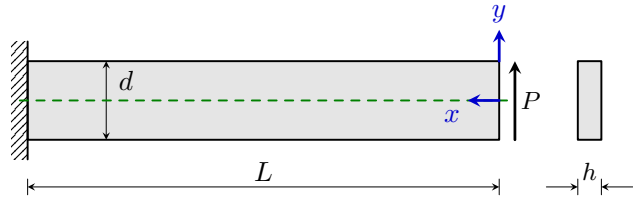
- Thus, the displacement fields are:

$$\begin{aligned}u(x, y) &= -\frac{3M}{2Eh^3}xy \\ v(x, y) &= \frac{3M}{4Eh^3}(x^2 + \nu y^2 - L^2)\end{aligned}$$



Bending of a Cantilever Beam

[Problem 2] : Derive the displacement, strain and stress fields for a cantilever beam loaded as shown considering it as a plane problem.



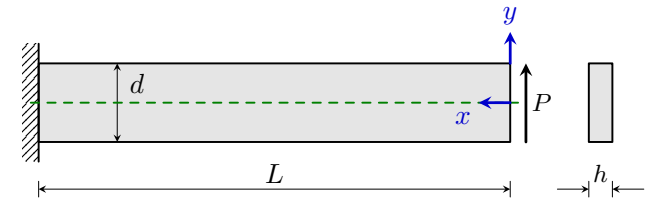
We Know that:

- Bending moment is proportional to the distance x from the point of application of load.
- Stress σ_{xx} is proportional to distance from the neutral axis y .

Therefore, $\sigma_{xx} = C_1xy = \phi_{,yy}$

Bending of a Cantilever Beam

[Problem 2] : Contd ...



- Step 1 : The Airy stress function for this case can be written as :

$$\sigma_{xx} = C_1xy = \phi_{,yy} \implies \phi_{,y} = \frac{1}{2}C_1xy^2 + f_1(x)$$

$$\phi = \frac{1}{6}C_1xy^3 + yf_1(x) + f_2(x)$$

Bending of a Cantilever Beam

[Problem 2] : Contd ...

- Step 2 : Determine $f_1(x)$ and $f_2(x)$ such that we know the form of the Airy stress function

$$\nabla^4 \phi = y \frac{d^4 f_1}{dx^4} + \frac{d^4 f_2}{dx^4} = 0$$

This equation has to be true for all values of x and y . Therefore,

$$\frac{d^4 f_1}{dx^4} = 0; \quad \frac{d^4 f_2}{dx^4} = 0$$

Integrating to obtain $f_1(x)$ and $f_2(x)$ will result in:

$$f_1(x) = C_2x^3 + C_3x^2 + C_4x + C_5; \quad f_2(x) = C_6x^3 + C_7x^2 + C_8x + C_9$$

Therefore, ϕ can be written as:

$$\phi(x, y) = \frac{1}{6}C_1xy^3 + y(C_2x^3 + C_3x^2 + C_4x + C_5) + C_6x^3 + C_7x^2 + C_8x + C_9$$

Bending of a Cantilever Beam

[Problem 2] : Contd ...

- Step 3 : Determine the stress field

$$\sigma_{xx} = \phi_{,yy} = C_1xy; \quad \sigma_{yy} = \phi_{,xx} = 6(C_2y + C_6)x + 2(C_3y + C_7)$$

$$\sigma_{xy} = -\phi_{,xy} = -\frac{1}{2}C_1y^2 - 3C_2x^2 - 2C_3x - C_4$$

Challenge : To determine the unknown coefficients.

- Step 4 : Apply boundary conditions

$$\sigma_{yy} = 0 @ y = \pm \frac{d}{2}; \quad \sigma_{xy} = 0 @ y = \pm \frac{d}{2}$$

Using the first boundary condition:

$$6 \left(C_2 \frac{d}{2} + C_6 \right) x + 2 \left(C_3 \frac{d}{2} + C_7 \right) = 0; \quad 6 \left(-C_2 \frac{d}{2} + C_6 \right) x + 2 \left(-C_3 \frac{d}{2} + C_7 \right) = 0$$

Bending of a Cantilever Beam

[Problem 2] : Contd ...

- **Step 4** : Contd ... Since these need to be true for all x , this results in $C_2 = C_3 = C_6 = C_7 = 0$. With these coefficients known the stress field becomes:

$$\sigma_{xx} = C_1xy; \quad \sigma_{yy} = 0; \quad \sigma_{xy} = -\frac{1}{2}C_1y^2 - C_4$$

To determine the unknowns C_1 and C_4 we use the second BC that results in $C_4 = -(1/8)C_1d^2$. Then:

$$\sigma_{xx} = C_1xy; \quad \sigma_{yy} = 0; \quad \sigma_{xy} = -\frac{1}{8}C_1(4y^2 - d^2)$$

Challenge : To determine C_1

Hint : We make use of the boundary condition that P is applied at $x = 0$.

Bending of a Cantilever Beam

[Problem 2] : Contd ...

- **Step 4** : Contd ... Using the boundary condition that P is applied at $x = 0$, we have:

$$\int_{-d/2}^{d/2} \sigma_{xy} h dy = -P \implies P = \frac{1}{8}C_1h \int_{-d/2}^{d/2} (4y^2 - d^2) dy = -\frac{1}{12}C_1hd^3$$

$$\implies C_1 = -\frac{12P}{hd^3} = -\frac{P}{I} \quad (I = \frac{1}{12}hd^3)$$

Thus the stress field is:

$$\sigma_{xx} = -\frac{Pxy}{I}; \quad \sigma_{yy} = 0; \quad \sigma_{xy} = -\frac{P}{2I} \left(\frac{d^2}{4} - y^2 \right)$$

Note: C_5, C_8 and C_9 in ϕ were not determined. As stresses do not depend on these constants, they are not needed and can be assumed to be zero.

Bending of a Cantilever Beam

[Problem 2] : Contd ...

- **Step 5** : Determine the strain field.

$$\epsilon_{xx} = \frac{\partial u}{\partial x} = \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{yy}}{E} = -\frac{P}{EI}xy$$

$$\epsilon_{yy} = \frac{\partial v}{\partial y} = \frac{\sigma_{yy}}{E} - \nu \frac{\sigma_{xx}}{E} = \frac{\nu P}{EI}xy$$

$$2\epsilon_{xy} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\sigma_{xy}}{G} = -\frac{(1+\nu)P}{EI} \left(\frac{d^2}{4} - y^2 \right)$$

- **Step 6** : Compute the displacement field by integrating the strains as:

$$u = -\frac{Px^2y}{2EI} + g_1(y); \quad v = \frac{\nu Px^2y}{2EI} + g_2(x)$$

To determine $g_1(y)$ and $g_2(x)$ we **use** the shear strain relation.

Bending of a Cantilever Beam

[Problem 2] : Contd ...

- **Step 6** : Contd ...

$$g_1'(y) - \frac{P}{EI} \left(1 + \frac{\nu}{2} \right) y^2 = -g_2'(x) + \frac{Px^2}{2EI} - \frac{(1+\nu)Pd^2}{4EI} = a_1$$

Solving we get:

$$g_1(y) = \frac{P}{3EI} \left(1 + \frac{\nu}{2} \right) y^3 + a_1y + a_2; \quad g_2(x) = \frac{Px^3}{6EI} - \frac{(1+\nu)Pd^2x}{4EI} - a_1x + a_3$$

Thus the displacement field takes the form:

$$u = -\frac{P}{2EI}x^2y + \frac{P}{3EI} \left(1 + \frac{\nu}{2} \right) y^3 + a_1y + a_2$$

$$v = \frac{\nu P}{2EI}xy^2 + \frac{P}{6EI}x^3 - \frac{(1+\nu)Pd^2}{4EI}x - a_1x + a_3$$

Bending of a Cantilever Beam

[Problem 2] : Contd ...

• **Step 6** : Contd ...

To determine a_1 , a_2 and a_3 we use the displacement boundary conditions, i.e.,

$$u(L, 0) = v(L, 0) = \frac{\partial v(L, 0)}{\partial x} = 0$$

Using these in the above relations result in:

$$a_1 = \frac{\nu PL^2}{2EI} - \frac{(1 + \nu)Pd^2}{4EI}; \quad a_2 = 0; \quad a_3 = \frac{PL^3}{3EI}$$

Thus the displacement field can be written as:

$$u = -\frac{P}{2EI}x^2y + \frac{P}{3EI}\left(1 + \frac{\nu}{2}\right)y^3 + \frac{P}{2EI}\left[L^2 - (1 + \nu)\frac{d^2}{2}\right]y$$

$$v = \frac{\nu Px^2y}{2EI} + \frac{Px^3}{6EI} - \frac{PL^2x}{2EI} + \frac{PL^3}{3EI}$$

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Bending of a Cantilever Beam

[Problem 2] : Contd ...

- We can compare the elasticity solution to beam theory solution by computing the deflection curve along the neutral axis at $x = 0$ and $y = 0$, we get:

$$v(0, 0) = \frac{PL^3}{3EI}$$

This corresponds to the maximum deflection. Also, if we take the partial differential of $v(x, y)$ with respect to x twice we get:

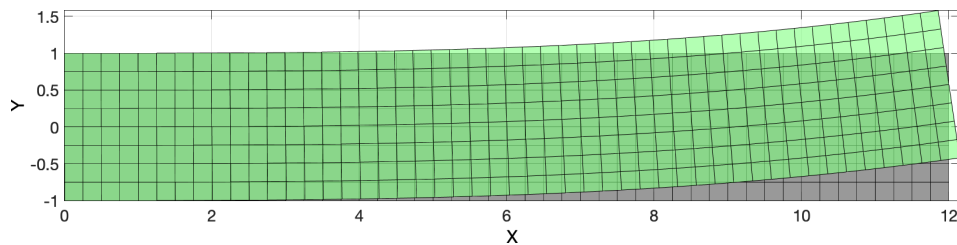
$$\frac{\partial^2 v}{\partial x^2} = \frac{Px}{EI} = \frac{M}{EI}$$

Note : This is the Euler-Bernoulli formula or moment-curvature relationship.

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Bending of a Cantilever Beam

[Problem 2] : Contd ...



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Working Assignment

Work out the plane stress and plane strain equations

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